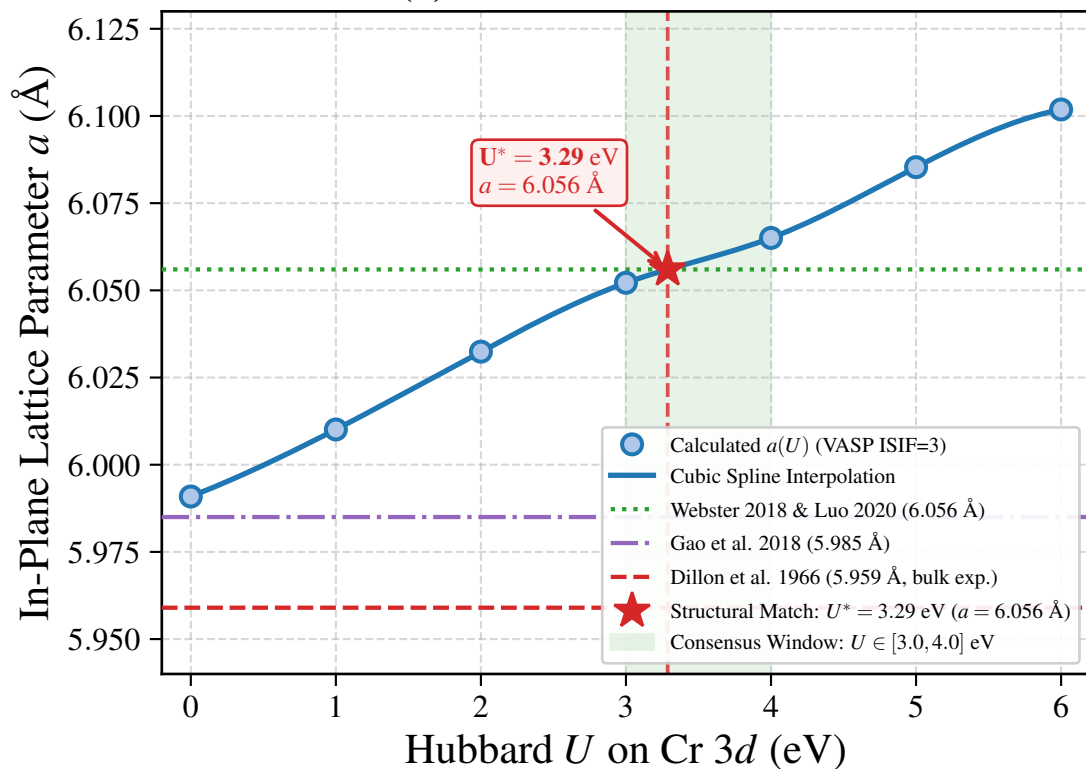
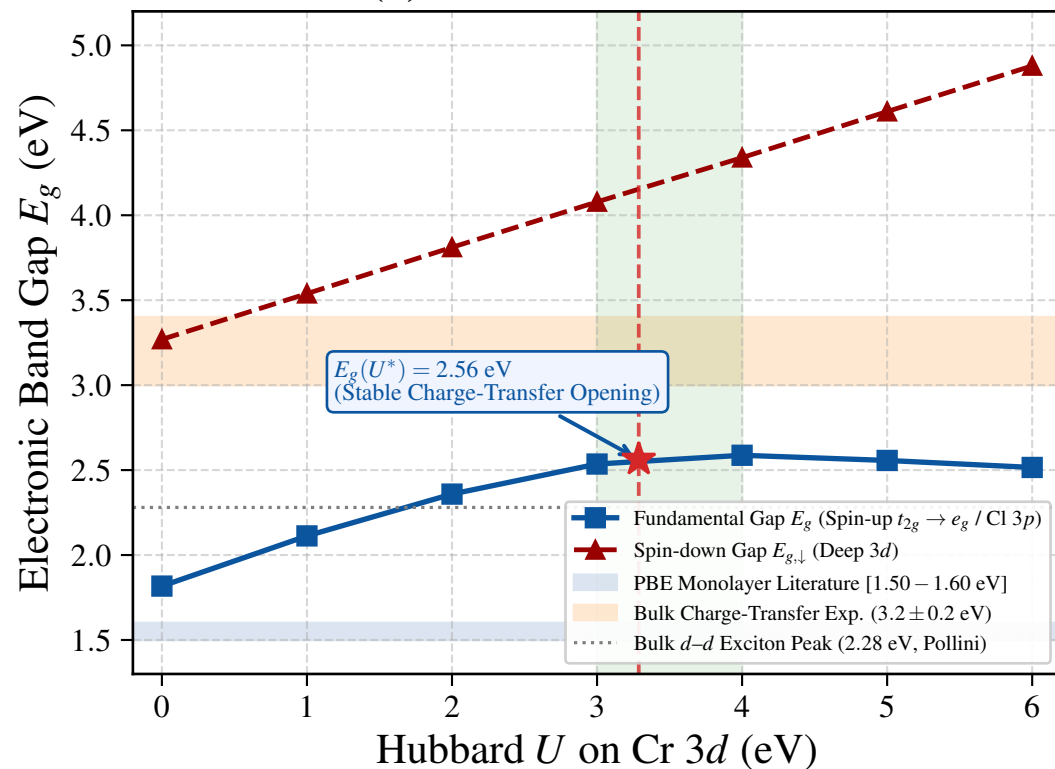


Monolayer CrCl₃: Optimal Hubbard U Determination via Multi-Constraint Optimization

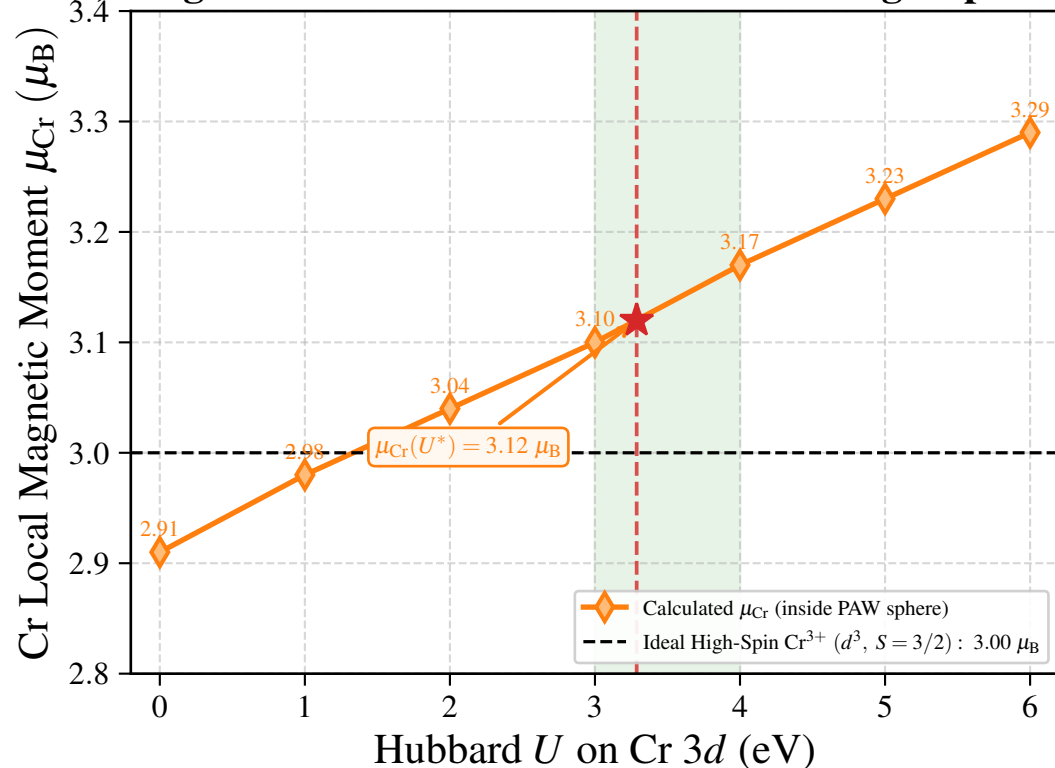
(a) Structural Constraint



(b) Electronic Constraint



(c) Magnetic Moment: Localization of 3d³ High-Spin State



Quantitative Synthesis & Recommendation for $2 \times 2 / 3 \times 3$:

1. Exact Structural Intersection with Webster 2018:

- $U^* = 3.29$ eV yields $a = 6.0560$ Å (0.00% error vs Webster 2018 & Luo 2020)
- $E_g(U^*) = 2.56$ eV (matches quasi-particle charge-transfer opening)
- $\mu_{Cr}(U^*) = 3.12 \mu_B$ (localized $S = 3/2$ high-spin magnetic state)

2. Production Recommendation for Fixed- U Calculations:

[Option A] $U = 3.0$ eV (Lowest Structural Bias)

→ $a = 6.052$ Å (−0.06% vs Webster), $E_g = 2.54$ eV, $\mu_{Cr} = 3.10 \mu_B$

[Option B] $U = 4.0$ eV (Standard Trihalide Benchmark)

→ $a = 6.065$ Å (+0.15% vs Webster), $E_g = 2.59$ eV (peak CT gap)

→ Cococcioni linear-response consensus for Cr 3d in halides (~ 3.8 eV)

→ Matches pre-staged benchmark sites (H_S1_U4, H_S2_U4)

3. Recommended Scientific Strategy for Manuscript:

- Compute full parametric series $U \in [0, 6]$ eV (already staged in crcl3-2x2)
- Present $U = 3.0$ eV or $U = 4.0$ eV as the primary baseline in the main text
- Report $U = 0 \dots 6$ eV as sensitivity analysis proving HER activity robustness!